

Visible Inequality and Voter Support for Public Goods: Evidence from Ohio School Finance Referenda*

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Abstract

Does inequality reduce willingness to collectively fund public goods? We examine this question using data from 3,273 school district referenda from the state of Ohio for the period from 2009 to 2019. We examine the differential effects of income inequality typically captured in economic surveys and the inequality that voters themselves perceive through variation in their community's housing stock. Our primary measure of perceived inequality is the 90/10 ratio of home values within a district — a dimension of inequality made salient by the physical landscape voters encounter in their everyday lives. We find that greater visible housing inequality is associated with a lower referendum yes-vote share whereas the Gini coefficient of income inequality is not. The effects of perceived inequality are concentrated in geographically compact districts where routine exposure to housing disparities is highest, and absent in more spread-out districts where such exposure is lower. The negative effect is also substantially attenuated in counties with stronger cross-income social ties, consistent with bridging social capital sustaining solidarity in the presence of salient inequality. Our results suggest that voters respond to the inequality they perceive in their community rather than inequalities in the underlying income distribution that are commonly used by social scientists to explain voter behavior.

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1 Introduction

A central question in political economy is whether inequality undermines the collective provision of public goods. The canonical Meltzer–Richard model predicts that as the income distribution becomes more skewed, the median voter’s income falls relative to the mean, and the demand for redistribution rises (Meltzer and Richard, 1981). However, alternative theories posit that increased heterogeneity reduces support for public spending (Romer et al., 1992; Balsdon et al., 2003). Perhaps unsurprisingly, empirical tests of this prediction have been mixed, but tend to suggest that inequality often reduces, rather than raises, support for public spending (Alesina and Giuliano, 2011). The mechanisms invoked to explain the negative pattern emphasize social fragmentation and in-group preferences: when communities are visibly stratified, voters identify less with the broader public for whom collective spending is intended, and the willingness to fund shared investments falls. We contribute to the broader question of the impact of inequality on the provision of public goods by bringing a new measure of inequality, specifically the within-district dispersion in home values, which is easily perceived, and find that higher levels of visible housing inequality reduce voter support for collective public-good investments.

A large empirical literature has tested the Meltzer–Richard prediction by relating various statistical measures of income inequality (the Gini coefficient, top income shares, and percentile ratios) to mass attitudes toward redistribution and to public-spending choices. The results have been inconsistent in sign and frequently null (Alesina and La Ferrara, 2005; Kelly and Enns, 2010; Kenworthy and McCall, 2008; although for a contrary view, see Boustan et al., 2013). However, a growing literature emphasizes that what shapes voters’ preferences over redistribution and public goods is not the statistical distribution of incomes, which most voters cannot accurately characterize, but the inequality they perceive in their environment (Gimpelson and Treisman, 2018; Hauser and Norton, 2017; Norton and Ariely, 2011). Survey evidence consistently shows that perceived inequality diverges from actual inequality and that it is those *perceptions*, as opposed to the underlying distribution, which predict policy preferences (Domènech-Arúmi, 2025; Suss, 2023). Thought-provoking as this literature is, a key limitation is its reliance on stated attitudes, often in the laboratory setting, rather than choices made in the real world with significant fiscal stakes.

We test these ideas in the context of school district referenda, which offers several distinct advantages. Referenda provide a direct, incentive-compatible measure of voter preferences over local public goods. Each observation is an actual vote, not a survey response, and involves a concrete, quantifiable fiscal decision: whether to authorize a new tax burden for the most prominent local

public good, K–12 public education. In contrast to many federal or even state-level redistributive programs where the flow of costs and benefits is diffuse and uncertain, the costs and benefits of these referenda are localized. The outcomes of the referenda we study directly impact voters, both in terms of costs (additional tax burden) and benefits (improved educational outcomes and increased property values; Baron, 2022; Cellini et al., 2010; Jackson et al., 2016). Additionally, we use referenda data from Ohio, a state that is politically and economically mixed and often exemplifies middle America. Our data span a long window from 2009 to 2019, encompassing the Great Recession, the post-recession recovery, as well as the pre-pandemic expansion, stopping just before the severe disruptions induced by the COVID-19 pandemic.

In this paper we focus on one dimension of inequality that is directly observable to voters: the disparity in home values within a school district. Our primary measure of inequality is the ratio of 90th percentile to 10th percentile of owner-occupied home values (“90/10 ratio”) within each school district, constructed from the American Community Survey. Voters acquire information about housing inequality through multiple channels. They see disparities in the physical landscape when they drive through their district on the way to work, the doctor’s office, or the grocery store. They are also able to encounter home-value information easily through online platforms such as Zillow and Realtor.com, where the values of specific properties are routinely searched. They also see the homes of other parents through their children’s friendships and school activities, gaining a social as well as a geographic view of the community’s housing stock. Each of these channels makes the home-value distribution visible. Incomes, by contrast, are private information, rarely disclosed outside of the household setting, and accordingly voters must form impressions of the income distribution from indirect cues rather than through direct observation. Indeed, similar measures of housing inequality have been operationalized and found to influence voter preferences for their elected official’s ideology (Brännlund and Szulkin, 2023; Song and Kang, 2025).

The main finding of our paper is that greater visible housing inequality is associated with a lower yes-vote share. The coefficient on the 90/10 ratio is negative and statistically significant across a sequence of specifications that progressively add controls for the Gini coefficient of income inequality, ethnic fractionalization, and an extensive set of socioeconomic and referendum characteristics. We find that a one-standard-deviation increase in the 90/10 ratio is associated with a reduction in the yes-vote share of approximately 0.68 percentage points in the most parsimonious specification and approximately 0.45 percentage points in the most demanding one. In contrast, the estimated effect of the Gini coefficient of income inequality is not statistically significant in the full model and changes signs across specifications. The contrast between the stable significance

of the home-value inequality and the lack of effect of the Gini coefficient is consistent with our view that what shapes voter behavior is the inequality they can directly observe, not the statistical income distribution they cannot.

We conduct multiple heterogeneity tests that bolster our interpretation and reveal critical insights into the main results. First, the negative effect of the 90/10 ratio is concentrated in geographically compact districts, defined as those below the sample median land area of approximately 48 square miles. In the compact-district subsample, the coefficient on the 90/10 ratio is roughly twice the magnitude of the full-sample estimate and statistically significant at the 5 percent level. In contrast, the coefficient on the 90/10 ratio is statistically indistinguishable from zero in the large-district subsample. This split is consistent with a visibility or perception mechanism. In compact districts, residents are likely to routinely encounter the full range of their community’s housing stock as part of their experience, and the contrast between high-value and low-value neighborhoods is likely to be more salient to them. In large districts, by contrast, geographic scale likely limits cross-neighborhood exposure and inhibits the ability of voters to form beliefs about the dispersion in values of the district’s housing stock.

Second, the effect of perceived inequality is moderated by the degree of cross-income social integration within the community. We measure this construct using the economic connectedness index constructed by Chetty et al. (2022) from Facebook friendship data, which captures the extent to which lower-income and higher-income residents form social ties with one another. The effects of house price inequality on the yes-vote share are larger in districts with lower levels of economic connectedness. In contrast, the negative effects of visible housing inequality are substantially attenuated in districts with higher levels of economic connectedness. Our interpretation of these results is that social capital that bridges between economic strata offsets the fragmentation associated with visible housing disparities and preserves collective willingness to fund public goods even when inequality is present.

In addition, both heterogeneity results mitigate concerns about our cross-sectional identification. An omitted variable correlated with the home-value 90/10 ratio and the yes-vote share could bias our results and drive the observed negative relationship. Instead, the inequality coefficient is concentrated in compact districts and in districts with low economic connectedness and absent in other districts. These findings are difficult to explain by a critical omitted variable, but are consistent with our proposed visibility mechanism: housing inequality is salient to voters in districts where the housing stock is observable, and the political response to that salience is attenuated in communities where cross-income social ties knit residents of different income levels together.

To the best of our knowledge, this paper is the first to bring together (i) direct measurement of observable inequality, as distinct from statistical income inequality, with (ii) a large dataset of actual voter choices over public goods provision, and (iii) mechanism tests that leverage both geographic and social capital variation to bolster and enrich the findings. We extend the perceptions-of-inequality literature by moving from survey and laboratory evidence (Choi, 2019; Gimpelson and Treisman, 2018; Hauser and Norton, 2017; Norton and Ariely, 2011) to revealed preference in a real-world fiscal setting. Similar to us, Balsdon et al. (2003) examine school referendum passage, but find income dispersion has a negative and statistically significant impact on yes-vote share, whereas housing price dispersion does not have a statistically significant impact, though the variables included in the model are constrained for theoretical reasons. We also connect to an extensive literature examining how socioeconomic characteristics and funding priorities affect public-school referendum passage (e.g., Bowers et al., 2010; Bowers and Lee, 2013; Lang and VanCeylon, 2025; Rubinfeld, 1977; Zimmer et al., 2011). In this context, voter psychology has been found to influence approval, as evidenced by ballot order effects, tax-word aversion, and status quo bias (Augenblick and Nicholson, 2016; Brunner et al., 2018; Lang and Ricchio, 2026), and our findings about perceived inequality are more related to this work.

The paper closest in spirit to ours is Boroomand and Kaul (2026). Like us, they take the perceptions-and-public-goods question out of the laboratory and into real-world voting data, but focus on racial diversity, rather than housing inequality. They find that the diversity penalty on local school spending and bond passage is concentrated in less inclusive communities, as measured by low residential integration and low economic connectedness. In racially diverse but socially and residentially integrated communities, the diversity penalty disappears. Adapting their approach, we consider the same two moderators and ask a parallel question for a different dimension of community fragmentation: visible disparities in housing values within a given school district; as we show subsequently, the economic-connectedness moderator carries through cleanly in our setting while the residential-integration moderator does not. The underlying intuition is shared across both papers: the social fabric of a community shapes its capacity for collective action in the face of salient differences, but the source of fragmentation differs. While Boroomand and Kaul examine differences in racial composition, we examine differences in economic standing that are visible through the built environment.

The remainder of the paper proceeds as follows. Section 2 describes the institutional context. Section 3 presents the data and empirical strategy. Section 4 reports the main results, heterogeneity analyses, and robustness checks. Section 5 concludes.

2 Institutional Background: School Finance in the United States

Elementary and secondary public education in the United States is financed jointly by federal, state, and local governments. In fiscal year 2023, the national breakdown was approximately 13 percent federal, 45 percent state, and 42 percent local (U.S. Census Bureau, 2025). The local share — close to that of the state — is financed primarily through property taxes, which ties school revenue directly to the residential real estate base of each district.

Ohio follows this broad pattern but is more local-heavy than the national average. In fiscal year 2024, Ohio school districts drew approximately 50 percent of revenues from local sources, primarily property taxes supplemented by a smaller local income tax component, with the state contributing roughly 37 percent and the federal government 13 percent (Signal Cleveland, 2025, drawing on Ohio Legislative Service Commission data). What makes Ohio’s reliance on local property taxation consequential is a feature of state law that is largely invisible to casual observers. The Ohio legislature passed House Bill 920 in 1976 to protect homeowners from rising tax bills during a period of rapid inflation in property values. The law instructs county auditors to apply a tax reduction factor to each voted property tax levy in their county, scaling the effective millage rate down each year so that the levy collects the same total dollar amount from existing property as it did the previous year. County auditors update assessed property values every six years through a full reappraisal, with a statistical update at the three-year midpoint. As assessed home values rise, the auditor’s reduction factor rises in parallel, and the district collects no additional revenue from the levy in nominal terms. The real value of that revenue therefore falls as inflation eats into it.¹

The result is that a district whose operating costs rise with inflation must return to its voters for permission to grow its operating revenue. School districts ask voters for two main kinds of approval. The first is a levy, a continuing tax approved at the ballot. Levies fund current operating expenses, including teacher salaries, instructional materials, and transportation, whose benefits flow to students currently enrolled. Some levies also fund capital improvements such as equipment

¹Several smaller revenue channels lie outside the HB 920 rollback. Bond levies and school emergency levies adjust automatically each year to raise the specific dollar amount that voters approved, so they generate stable revenue without the rollback kicking in. New construction also adds to the tax base in its first year before the rollback absorbs it in subsequent years. School districts also receive a share of “inside millage” — a small pool of unvoted property tax authority granted by the Ohio constitution to local governments collectively, with a cap of ten mills across all overlapping local entities. Because inside millage is unvoted, HB 920 does not apply to it. Finally, the rollback’s reduction factor on voted operating millage cannot scale the effective rate below twenty mills. Districts at or near that floor receive some real revenue growth from rising property values; districts above the floor experience the full rollback. These exceptions allow modest revenue growth at the margins, but they do not change the basic institutional dynamic that school districts are often left with little choice but to go back to their voters for authorizing spending.

purchases and building repairs that fall short of full new construction. The second instrument is a bond, which authorizes the district to borrow against future tax revenues to finance large capital projects such as school construction, renovation, and major facility upgrades. Before a bond measure can appear on the ballot, it must be certified by the Ohio School Facilities Commission. A simple majority of votes cast is sufficient for passage of both bonds and levies. With over one trillion dollars in school bond debt outstanding nationally, approximately \$700 billion of which originates in states that require voter approval, school district referenda constitute one of the largest arenas of direct democracy in the American fiscal system.

The bond-versus-levy distinction is institutional, but for our analysis the more relevant distinction is between the purpose of the spending: capital expenditures versus current operations.² Capital projects finance durable, location-specific investments such as a new high school, a renovated middle school, or a new athletic facility. The benefits of any single project accrue over decades, but they accrue most directly to the students and families served by that particular building. Operating expenditures, by contrast, fund current services whose benefits flow to all enrolled students across the district: teacher salaries, instructional materials, transportation, and the like. The economic distinction is consequential for our hypothesis. A voter in a district with visible housing inequality has reason to ask whether a new school facility will be built in their neighborhood or in a wealthier part of the district, and whether their children will still be enrolled in the public school system when it opens. Capital projects also commit voters to a longer repayment horizon, up to 40 years for some bond measures, raising the stakes of any distributional skepticism. In contrast, operating expenditures raise this question less acutely because their benefits flow more evenly across the district and they can commence more immediately as a district hires more teachers or more counselors or enacts salary increases for its staff. We test in Subsection 4.5 (rows R4a and R4b) whether the visibility-of-inequality effect on the yes-vote share is more pronounced for referenda involving capital expenditures than for operating expenses.

²The bond-versus-levy distinction is correlated with the capital-versus-operating distinction but does not coincide with it. Bonds finance capital projects, but some levies also fund capital purposes such as equipment purchases or building repairs that fall short of full new construction. For the heterogeneity analysis in Subsection 4.5, we classify referenda by spending purpose using the *text* of each ballot measure rather than by the financing instrument.

3 Data and Empirical Strategy

3.1 Data

Referenda outcomes. Referendum outcome data come from the Ohio Secretary of State’s official election results. The outcome variable in all specifications is the yes-vote share, defined as the percentage of yes votes among all valid votes cast: $\text{Yes-vote share} = 100 \times \text{yes} / (\text{yes} + \text{no})$. Our sample spans 2009 to 2019. The raw data include all bond and levy referenda placed before voters by Ohio school districts during the sample period. We exclude observations of three types. First, we drop three measures that propose a reduction in an existing tax rate rather than authorize new revenue, since the political economy of approving a tax cut differs from that of approving a tax increase and is outside the scope of our research question. Second, prior to 2012 the Secretary of State’s office published bond-plus-levy packages across separate documents, producing multiple rows in the raw data for a single ballot question. We identify 38 such referendum packages over the sample period. For each package, we retain one observation, the bond, and reclassify the spending purpose as mixed to reflect the combined fiscal nature of the question. Third, we drop one duplicate-millage row from a 2010 ballot that is a clerical artifact rather than a distinct measure. After these exclusions, our final sample comprises 3,273 referenda.

Referenda appear on the ballot during what the Ohio Secretary of State classifies as general, primary, or special elections. By that classification, general elections are held in November and account for 54 percent of the sample (1,753 observations), primary elections are typically held in May and account for 40 percent (1,311 observations), and special elections are held either in August or February and account for the remaining 6 percent (209 observations).

For our empirical analysis, however, we use a coarser two-way split that groups referenda by the type of electorate likely to turn out at the polls. We define a general-election referendum as one held in November of an even-numbered year, when federal and gubernatorial contests also appear on the ballot. The remaining referenda (primary, special, and November elections in odd-numbered years) are all grouped together under a non-general label. The motivation for this regrouping is that even-year November general elections attract a broader and more representative electorate than the other settings, in which turnout tends to be lower and skewed toward more engaged voters (Hajnal and Lewis, 2003; Kogan et al., 2018). Under this regrouping, general-election referenda account for 858 observations (26 percent of the sample), and non-general referenda account for the remaining 2,415 observations (74 percent).

Finally, we classify each referendum by its spending purpose using the text of the ballot mea-

sure as recorded by the Secretary of State. Measures that propose new construction, renovation, or major facility upgrades are classified as capital. Measures that propose funding for teacher salaries, instructional materials, transportation, or other current operating expenses are classified as operating. A residual category labeled mixed captures referenda whose ballot text bundles capital and operating purposes into a single vote. Operating referenda are by far the largest category in our sample, accounting for 67.6 percent of observations (2,213 referenda). Capital referenda account for 27.7 percent (905), and mixed referenda for the remaining 4.7 percent (155). Not only are voters likely to favor certain types of funding more (e.g., Zimmer et al., 2011), but perceived inequality may also interact with funding type to influence approval.

Visible housing inequality. Our primary explanatory variable is the 90/10 ratio of owner-occupied home values within each school district, $HV_{90/10}$, constructed from the American Community Survey (ACS). The ACS reports the proportion of homes falling into various bins, and we construct the 10th and 90th percentile estimates using linear interpolation within each bin.³ A ratio of 1.0 would indicate perfect equality between the 90th-percentile home and the 10th-percentile home in the district. Higher values indicate greater dispersion. The sample mean of $HV_{90/10}$ is approximately 5.7, with substantial variation across districts. To prevent a small number of extreme observations from driving the results, we winsorize $HV_{90/10}$ at the 1st and 99th percentiles. The winsorized variable is what enters all of our regressions.

ACS vintage and merge. The ACS releases five-year estimates each year for every school district, with the estimate labeled “year t ” covering the five-year window from $t - 4$ through t . We use three largely non-overlapping vintages of the five-year ACS, the 2005–2009 release, the 2008–2012 release, and the 2012–2016 release, and merge each vintage to referenda whose election year falls within a defined lagged window. Specifically, we match referenda in 2009 through 2012 to the 2005–2009 ACS, referenda in 2013 through 2016 to the 2008–2012 ACS, and referenda in 2017 through 2019 to the 2012–2016 ACS. This lagged assignment ensures that the ACS measure for each referendum reflects district demographics established before the vote, rather than contemporaneously with it. We assess the sensitivity of our findings to this merge in Subsection 4.5, where we also report results using an annually lagged variant and a predetermined single-vintage variant in which all referenda are merged to the 2005–2009 ACS, the first 5-year ACS conducted by the U.S. Census Bureau.

³One concern about using ACS data to construct $HV_{90/10}$ is that the Census Bureau top-codes owner-occupied home values at \$1,000,000. In our sample, no school district in any year has a 90th-percentile home value at or above this top-code; the highest district-year observation is approximately \$930,000. Top-coding therefore does not affect the construction of $HV_{90/10}$ for any observation in our regressions.

Controls. The full set of controls follows the perceived-inequality and school-finance literatures. From the ACS, we include the Gini coefficient of household income, ethnic fractionalization computed following Alesina et al. (2003), the natural logarithm of mean home value, the natural logarithm of median household income, the natural logarithm of population density, the homeownership rate, the fraction of residents twenty-five and older with a bachelor’s degree, the share of population under 18, and the share of population 65 and older. We include mean home value as a proxy for the average tax burden, which can be a significant determinant of willingness to take on additional taxes (Rubinfeld, 1977; BalsaDon et al., 2003). The prediction is that the yes-vote share will be negatively correlated with mean home value. We also include two additional controls: the natural logarithm of district enrollment in the prior school year (from the National Center for Education Statistics) and the county-level Democratic share of the two-party presidential vote in 2012, constructed using Dave Leip’s Atlas of U.S. Presidential Elections. In our fifth and final specification, we introduce categorical indicators describing the referendum itself: a set of dummies for the spending category (capital, operations, or mixed) and an indicator for non-general election.

Mechanism variables. For the heterogeneity analyses, we use three community-structure variables. The first is district land area in square miles, from the ACS. We define geographically compact districts as those below the sample median of approximately 48 square miles. The second is the economic connectedness index of Chetty et al. (2022), measured at the county level and constructed from Facebook friendship data. The index captures the extent of cross-income social ties in a community, with higher values indicating more social integration across income groups. The third, included to mirror the moderators studied in Boroomand and Kaul (2026), is the residential integration measure constructed as one minus Theil’s H computed across census tracts within each district. Higher values indicate more residential mixing of households of different incomes across the district’s tracts. Together these three variables let us test whether the visibility-of-inequality effect varies with the geographic scale of the district, the social integration of its residents across income lines, and the spatial integration of its housing stock.

Summary statistics. Table 1 reports summary statistics for the variables used in the analysis. The dependent variable is the yes-vote share, with a sample mean of approximately 54 percent and a standard deviation of 11 percentage points. The (un-winsorized) home-value 90/10 ratio has a sample mean of 5.7 and a standard deviation of 3.2, with a long upper tail: the median is 4.8 and the 99th percentile sits above 20. To account for this considerable dispersion, we winsorize the variable at the 1st and 99th percentile levels and that variable has a range between 2.7 and 20.5, as opposed to an earlier range for the unwinsorized variable of 2.4–35.9. District land

area averages about 62 square miles but is also right-skewed, with a median of 48 square miles. County-level economic connectedness ranges from roughly 0.6 to just above 1.0 across our sample, while residential integration sits between about 0.6 and 1.0. The remaining variables include standard demographic, economic, political, and referendum controls drawn from the ACS, the 2012 presidential election returns, and the Ohio Secretary of State.

Correlations among the inequality measures and moderators. Appendix Table A.1 reports pairwise correlations among the home-value 90/10 ratio, the Gini coefficient of household income, ethnic fractionalization, mean home values, district size (the compact-district indicator), county-level economic connectedness, and residential integration. Two patterns stand out. First, the home-value 90/10 ratio is only weakly correlated with the Gini coefficient, which is consistent with our argument that visible housing inequality and the statistical income distribution capture distinct features of a community. Second, the three moderators we use in the heterogeneity analysis — district size, economic connectedness, and residential integration — are themselves only modestly correlated with one another, so the split-sample tests for any one moderator are unlikely to be picking up the influence of another; Subsection 4.4 confirms this independence directly by entering the district-size and economic-connectedness interactions jointly.

3.2 Empirical Strategy

We estimate linear specifications of the form:

$$\text{Yes-vote share}_{irt} = \alpha + \beta \cdot \text{HV}_{90/10,it} + \mathbf{X}'_{irt}\gamma + \delta_t + \varepsilon_{irt}$$

where i indexes school districts, r indexes referenda, and t indexes election years. The variable $\text{HV}_{90/10,it}$ is the winsorized home-value 90/10 ratio for district i in the ACS vintage assigned to year t , standardized to a z-score so that the coefficient β reads directly as the effect on the yes-vote share of a one-standard-deviation increase in visible housing inequality. The vector $\mathbf{X}_{irt}$ collects the controls described in Subsection 3.1. The year fixed effects δ_t absorb common aggregate shocks: business-cycle fluctuations, shifts in the state political environment, and changes in the fiscal landscape that affect all districts. Standard errors are clustered at the school district level. The identifying variation comes from within-year, cross-district differences in visible housing inequality.

The main threat to identification is that $\text{HV}_{90/10}$ may be correlated with unobserved district characteristics that independently predict referendum outcomes. Our primary defense is the in-

clusion of an extensive set of controls including income level, educational attainment, ethnic fractionalization, age distribution, student enrollment, and homeownership that capture the most plausible confounders. Our second defense leverages heterogeneity analyses. If the $HV_{90/10}$ effect were driven by some unobserved district characteristic rather than by the visibility of housing disparities, we would expect it to operate similarly across districts regardless of their geographic scale or the social ties among their residents. Instead, as we show in Subsection 4.2, the effect is concentrated in geographically compact districts where the housing stock is more visible to voters, and it is substantially attenuated in counties with high economic connectedness, where cross-income social ties offset the fragmentation associated with visible inequality. In each case the moderation runs in the direction the mechanism predicts. A confounder that mimics this pattern would have to be systematically correlated with both district size and community connectedness in exactly the right way, which strikes us as an unlikely coincidence.

4 Results

4.1 Main Results

Table 2 presents our main estimates. The coefficient on the standardized home-value 90/10 ratio is negative and statistically significant (at the 10 percent level at least) across all five specifications. In Column 1, which includes only the home-value ratio and year fixed effects, the estimate is -0.68 (significant at the 5 percent level). Column 2 adds two controls drawn from the literature on provision of public goods — the Gini coefficient of household income inequality and ethnic fractionalization — and yields a coefficient of -0.75 , again significant at the 5 percent level. Column 3 adds the logarithm of mean home values, which lets us separate the level of housing wealth from its dispersion within the district, and yields a coefficient of -0.65 , again significant at the 5 percent level. Column 4 now introduces an extensive battery of demographic and economic controls and produces a somewhat attenuated estimate of -0.46 that is significant at the 10 percent level. Lastly, Column 5 adds indicators for the spending category of the referendum (capital, operations, or mixed) and for whether the referendum is held in a non-general election; the estimate is essentially unchanged at -0.45 and it continues to be significant at the 10 percent level.⁴ Thus, the

⁴One concern about the cross-sectional identification of this coefficient is that time-invariant district characteristics correlated with the housing-value distribution could be driving the negative association. We re-estimated all specifications reported in this table adding school-district fixed effects, which restricts identification to within-district variation in the home-value 90/10 ratio across the three ACS vintages. The coefficient on the home-value 90/10 ratio varies in a tight band between -0.51 and -0.56 across all five specifications, although it is less precisely estimated given the limited within-district variation. The persistence of the point estimate under district fixed effects argues against an interpre-

coefficient on perceived inequality, as captured by the standardized home-value 90/10 ratio, retains its sign and statistical significance from the most parsimonious to the most demanding specification, and although it attenuates moderately as controls are added, it does not collapse. The results in this table indicate that variation in visible housing inequality that predicts voter behavior is not absorbed by the standard controls that social scientists have used to model school referendum outcomes. In our subsequent discussion, we refer to the fifth column as our full specification.⁵

In terms of magnitude, the impact of perceived inequality is modest but meaningful. The (winsorized) home-value 90/10 ratio has a sample mean of 5.7 and a standard deviation of approximately 3, and we standardize it to a z-score so that the coefficient reads directly as the effect of a one-standard-deviation increase in visible housing inequality. A one-standard-deviation increase in the home-value 90/10 ratio is associated with a 0.45 percentage point reduction in the yes-vote share in our most demanding specification, and with a 0.68 percentage point reduction in the most parsimonious one. For context, the mean yes-vote share in our sample is approximately 54 percent, and roughly 36 percent of all referenda in our sample fail to pass. Thus the effect size, although a modest fraction of the cross-district variation in voting behavior, is precisely the kind of variation that can make a difference to whether a closely contested referendum passes or not.

In contrast to the stability in sign and significance on the house price dispersion measure, the Gini coefficient of statistical income inequality is not robustly significant. In Column 2, where it appears alongside ethnic fractionalization and year fixed effects, its coefficient is positive and significant at the 1 percent level. However, once the full set of demographic, economic, and political controls is added in Column 4, the coefficient turns negative and is statistically indistinguishable from zero, and it continues to be negative in the final specification. The fact that the coefficient on the Gini is fragile is consistent with our view that what shapes voter behavior is the inequality voters can directly observe in their environment, not the statistical income distribution that they cannot.

Among the control variables in the full specification, several are individually significant and

tation in which the headline result is driven by time-invariant district characteristics that the standard controls do not fully capture.

⁵One concern is that the negative association between visible housing inequality and the yes-vote share could be driven by sorting into private schools — that families who opt out of public schools vote against public-school funding in districts where visible housing inequality is highest. We re-estimated the column progression of Table 2 adding the K-12 private-school attendance rate (drawn from the ACS) as an additional control. The coefficient on the home-value 90/10 ratio is essentially unchanged across columns. The coefficient on the private-school attendance rate is negative, as expected, but not statistically significant in any column; the closest it comes is $p = 0.113$ in the full specification that includes all demographic, political, and election-related controls. The estimated effect implies that a one-standard-deviation increase in private school attendance is associated with a reduction in the yes-vote share of approximately half a percentage point.

worth discussion. The share of population under age 18 is positively related to referendum approval. Surprisingly, none of the studies of school referendum approval we reviewed includes this variable, but individual surveys find that voters with children are more likely to approve (Lang and VanCeylon, 2025; Rubinfeld, 1977). The fraction of residents 25 and older with a bachelor’s degree carries a strongly positive and precisely estimated coefficient, consistent with the well-established finding that more educated individuals and communities reliably support school investment (Bowers et al., 2010; Rubinfeld, 1977). The county-level Democratic share of the 2012 presidential vote is negative and significant, suggesting that Ohio school referenda perform worse in counties with a more Democratic electorate over our sample period. While this result is counter-intuitive given findings based on individual voters (Lang and VanCeylon, 2025; Rubinfeld, 1977), studies using aggregated data often find negative or neutral effects of Democratic vote share. We, however, treat this result with caution because political orientation is only measured at the county level (given how the underlying data are collected) rather than at the school district level. Ohio counties contain on average about seven school districts, and thus we miss any within-county variation in political orientation across districts in our measure. Finally, the logarithm of district enrollment is also negative and significant, indicating that larger districts have lower yes-vote shares on average. Several explanations are consistent with this pattern, including greater preference heterogeneity among voters in larger districts (as argued by Romer et al. (1992) and Balsdon et al. (2003)), weaker information flows, and demographic differences between large and small districts that the other controls do not fully absorb. The results indicate that mixed-purpose referenda receive 4.3 percentage points less approval than referenda focused on new construction and renovation, whereas operating referenda are statistically indistinguishable from the latter. Lastly, log of mean home value and population density are marginally significant. The coefficient on log of mean home value is negative, as expected.⁶ The coefficient on population density is positive, which is consistent with prior findings (Bowers et al., 2010; Brunner et al., 2018). The remaining variables are not statistically distinguishable from zero, and thus we find no evidence that they influence approval.

⁶One may be surprised that the coefficient on mean home value is only statistically significant at the 10 percent level, given the variable’s status as proxy for tax burden. A likely confounding factor is that school district expenditure choices, intergovernmental aid flows, and the property tax bill a voter faces all co-move with the level of district property wealth. The choice between mean and median home value turns out to be immaterial here. Substituting the log of median home value — which is less sensitive to a handful of high-value properties and is arguably the more natural proxy for the tax burden facing the median voter — reproduces the same pattern: a positive, insignificant coefficient in the baseline that turns negative and significant at the 10 percent level under the full specification, while leaving the home-value 90/10 coefficient essentially unchanged. Those additional results are available on request.

4.2 Heterogeneity by District Size

Our first heterogeneity test asks whether the negative effect of visible housing inequality on the yes-vote share is concentrated in districts where the housing stock is most visible to voters. To operationalize this notion, we split the sample at the median district land area (approximately 48 sq. miles) into geographically compact districts (those with area below the median) and large districts (area at or above the median). The visibility argument we have laid out in the paper predicts a stronger effect in compact districts, where residents routinely traverse the full range of the community’s housing stock, than in large districts, where geographic scale, perhaps amplified by any residential sorting, limits cross-neighborhood exposure.

Before turning to the regression estimates, we visualize the underlying relationship using two complementary figures. Figure 1 presents a binscatter of the residualized yes-vote share against the home-value 90/10 ratio, separately for compact and large districts. The negative slope in the compact subsample is clearly visible,⁷ whereas the large-subsample slope is essentially flat. Figure 2 presents the corresponding continuous picture, plotting the estimated marginal effect of the home-value 90/10 ratio on the yes-vote share as a function of district land area. The marginal effect is most negative for the smallest districts and weakens steadily as land area increases, with the point estimate crossing zero in the upper portion of the area distribution. Taken together the two figures support the interpretation that the median split we present in Table 3 is not an artifact of where we cut the land-area distribution: the negative effect of visible inequality is concentrated in geographically compact districts, fades in districts of intermediate size, and is statistically absent in the largest districts.

Table 3 now reports the regression estimates. All three columns use the full set of demographic, economic, and political controls along with the spending-category and election-type indicators. Column 1 presents the home-value 90/10 coefficient estimated for the compact-district subsample. The coefficient is -0.99 and is statistically significant at the 5 percent level, with a magnitude approximately twice the full-sample headline result reported in Table 2. Column 2 presents the parallel estimate for the large-district subsample. The coefficient is essentially zero ($+0.05$) and statistically indistinguishable from zero. Column 3 reports a fully interacted pooled specification in which every control is allowed to differ across compact and large districts, so the coefficient on the interaction of the home-value 90/10 ratio with an indicator for compact districts is the cross-subsample difference in the home-value 90/10 slope without imposing any common-

⁷Appendix Figure A.1 illustrates the robustness of the relationship for small districts to winsorizing the housing inequality more aggressively, at the 5th/95th percentiles, or at the 10th/90th percentiles.

control restriction. The interaction is -1.08 and statistically significant at the 5 percent level, confirming that the cross-subsample difference in slopes survives even after permitting heterogeneity in every other coefficient as well.

While these heterogeneity results primarily add nuance to our findings and bolster our intuition about how inequality dampens referendum passage, they also serve as a central piece of evidence against an omitted-variable interpretation of our main results. A confounder that is correlated with both the home-value 90/10 ratio and the yes-vote share, but operates through a channel other than visibility, would be expected to depress yes-vote shares similarly across compact and large districts at comparable levels of inequality. Instead, the inequality coefficient is concentrated in compact districts and is absent in large ones; the cross-subsample equality of slopes is rejected at the 5 percent level in the most flexible specification. The visibility mechanism explains this pattern naturally, and we are not aware of an alternative explanation that would predict it as readily.

In passing, we also note that the Gini coefficient is statistically insignificant in both compact and large districts for the full specification that includes all controls. In fact, the coefficient flips sign across the two subsamples, positive in compact districts and negative in large ones, further supporting our view that income inequality plays a limited role in explaining voter referenda patterns.

4.3 Mechanism: Social Connectedness

While Subsection 4.2 establishes that visibility is the operative channel that drives the negative effect of housing inequality on the yes-vote share, we further investigate which factors influence whether that salience translates into political opposition to shared fiscal investment. We propose one candidate mechanism: the degree to which residents of different income levels form social ties with one another. Communities in which lower-income and higher-income residents are knit together by friendships and routine social interaction may sustain the cooperative norms that support collective spending, even when visible disparities in housing values make inequality salient otherwise. Communities in which such ties are weak may be more susceptible to the fragmentation that visible inequality would otherwise produce.

To test this hypothesis, we use the economic connectedness index of Chetty et al. (2022), constructed at the county level from Facebook friendship data. The index measures the share of high-income friends among low-income users in each county, capturing the extent to which lower-

and higher-income residents form social ties across the income distribution. We split the sample at the median of this index into low-connectedness and high-connectedness counties. Figures 3 and 4 visualize the moderating role of economic connectedness (EC). Figure 3 presents a binscatter of the residualized yes-vote share against the home-value 90/10 ratio, separately for districts in low-EC and high-EC counties. The negative slope in the low-EC panel is pronounced and visibly steeper, while the high-EC panel shows essentially no relationship between visible housing inequality and yes-vote share.⁸ Figure 4 presents the continuous version of the same result, plotting the estimated marginal effect of the home-value 90/10 ratio on the yes-vote share as a function of community connectedness. The marginal effect is sharply negative at the low end of the connectedness distribution, narrows steadily as connectedness increases, and crosses zero at approximately 0.86 on the connectedness scale. This crossing point corresponds to the 74th percentile in the distribution of economic connectedness across Ohio counties; thus the negative effect of visible inequality is statistically distinguishable from zero across roughly the lower three quarters of the distribution, and is absent in the upper quarter. The two figures jointly support the proposed mechanism: the negative effect of visible housing inequality is concentrated in communities where cross-income social ties are weak, and absent where they are strong.

Table 4 Panel A reports the corresponding regression estimates. All three columns use the full set of demographic, economic, and political controls along with the spending-category and election-type indicators. Column 1 presents the home-value 90/10 coefficient in the below median economic connectedness (low-EC) subsample. The coefficient is -1.37 and is statistically significant at the 1 percent level, with a magnitude roughly thrice the size of the pooled headline estimate in Table 2. Column 2 presents the parallel estimate for the high-EC (above median) subsample. The coefficient is small, positive ($+0.40$), and statistically indistinguishable from zero. Column 3 reports a fully interacted pooled specification that mirrors the structure of Column 3 in Table 3. We define a binary high-EC indicator that equals one for counties at or above the median of the economic connectedness distribution, and we allow every control to vary across the high- and low-EC groups. The coefficient on the home-value 90/10 ratio (-1.34) recovers the slope in low-EC counties and aligns closely with the Column 1 split-sample estimate of -1.37 . The interaction of the home-value 90/10 ratio with the high-EC indicator is positive ($+1.66$) and statistically significant at the 1 percent level, indicating that the negative effect of housing inequality is completely offset in counties where cross-income social ties are stronger. The continuous-EC analog of this pattern,

⁸Appendix Figure A.2 illustrates the robustness of the relationship for low EC counties to winsorizing the housing inequality more aggressively, at the 5th/95th percentiles, or at the 10th/90th percentiles.

shown in Figure 4, traces how the marginal effect of the home-value 90/10 ratio narrows steadily as economic connectedness rises.

The economic connectedness heterogeneity adds force to the identification arguments we develop in Subsection 4.2. Either the district-size or the economic-connectedness heterogeneity pattern in isolation could reflect a confounder correlated with the relevant moderator. To produce both patterns at once, however, a confounder would have to be correlated with district size and with county-level economic connectedness, both in the right direction. Visibility delivers both patterns through a single mechanism: housing inequality is salient to voters in districts where the housing stock is observable, and the political response to that salience is attenuated in communities where cross-income social ties knit residents of different income levels together.

Boroomand and Kaul (2026), whose paper is closest in spirit to ours, study two community-structure moderators of the diversity penalty on local school spending: economic connectedness and residential integration. The latter measures the extent to which households of different incomes live in the same neighborhoods within a school district, with higher values indicating that low-income, middle-income, and high-income residents are more spatially mixed across census tracts rather than sorted into income-homogeneous tracts.⁹ They find that the diversity penalty disappears when either moderator is high. We adopt both moderators and ask the parallel question for visible housing inequality rather than racial diversity. The economic connectedness result reported above is the strong analog of their finding. The pattern with the residential integration measure however is different: as Panel B of Table 4 illustrates, an interaction of the home-value 90/10 ratio with residential integration is statistically indistinguishable from zero in the fully interacted pooled specification. Moreover, the low- and high-integration subsample coefficients are not meaningfully different from each other either.

The asymmetry between the two moderators is itself substantively interesting. It suggests that, for the visibility-of-housing-inequality channel we examine in this paper, what matters is whether residents of different income levels form social ties with one another, not whether they live in spatially mixed neighborhoods within the district. Spatial proximity across income lines is apparently not sufficient on its own to attenuate the inequality effect; the social ties that might accompany proximity, but do not always do so, are what moderate the impact of housing inequality.

⁹Residential integration is computed as one minus Theil's H across census tracts within each district. See Boroomand and Kaul (2026) for details on the construction.

4.4 Joint Moderation by District Size and Economic Connectedness

Sections 4.2 and 4.3 establish that district size and economic connectedness each moderate the home-value 90/10 effect when entered one at a time. A natural follow-up question is whether the two moderators are independent, so that each captures a distinct dimension of community structure, or whether one largely subsumes the other when both are allowed to compete within a single specification.

Table 5 repeats the column progression from Table 2 but adds both interactions to every specification. Each control present in a given column is fully interacted with both moderators, so the two interaction coefficients can be read as the cross-size and cross-EC slope differences allowing the controls themselves to vary with each moderator. The interaction of the home-value 90/10 ratio with the high-EC indicator is positive across all five columns and statistically significant at conventional levels throughout, achieving significance at the 1 percent level in Columns 4 and 5. The interaction with the small-district indicator is negative in every column; it is significant at the 5 percent level in Column 2 but not statistically significant in the remaining four columns, with p-values ranging from 0.11 to 0.16. Both moderators retain the sign we would expect from the perceptions-and-social-ties channel when entered jointly, but the high-EC interaction retains significance in every column whereas the small-district interaction does not. We read this pattern as evidence that each dimension of heterogeneity we examine carries independent moderating power for the visibility-of-housing-inequality channel, but the social-ties dimension captured by economic connectedness carries the heavier weight in the joint specification.

4.5 Robustness

We assess the robustness of the headline result along six dimensions: stability across subperiods, the part of the housing-value distribution that drives the effect, the choice of ACS-to-referendum merge, the spending purpose of the referendum, the election type, and the treatment of districts that vote on more than one referendum within a tight time window. Table 6 summarizes the exercise in compact form. Each cell reports the home-value 90/10 coefficient under the Full specification (column 5 of Table 2) for a given robustness row and one of three samples: the overall sample (column 1), the compact-district only subsample (column 2), and the subsample of districts in low-EC counties (column 3). Reading down a column traces how the headline coefficient moves as we vary the robustness check within a fixed sample; reading across a row asks whether the two heterogeneity-based subsamples — in which our mechanism predicts the effect should be sharper

— continue to deliver larger coefficients than the overall sample under each robustness variant. The full five-column control progression underlying each cell (going from the most parsimonious specification to the Full specification) appears in three companion tables: Appendix Table A.2 for the overall sample, Table A.3 for the compact-district only subsample, and Table A.4 for the low-EC subsample.

Two patterns are visible at a glance. First, the home-value 90/10 coefficient is generally larger in absolute value in the compact and low-EC subsamples than in the overall sample, with magnitudes typically 1.5 to 3 times the corresponding overall-sample value. The baseline R0 row reads -0.45 , -0.99 , and -1.37 across the three samples, all significant at conventional levels, and the same relative ordering holds for the large majority of the robustness rows below it. Second, the rows in which the overall-sample coefficient attenuates to insignificance (R1a, R3b, R4b, R5a, R5b) are generally rows in which the compact and low-EC coefficients remain meaningful, suggesting that any attenuation we observe in the pooled sample reflects the dilution of a real effect by districts where the visibility mechanism is muted rather than the absence of an effect altogether. We work through each block of rows in turn, citing Table 6 for the cross-sample magnitudes and Appendix Table A.2 for the column progression within the overall sample.

Subperiod stability. Rows R1a and R1b of Table 6 split the sample into an early subperiod (2009 through 2013; $N=1,801$) and a late subperiod (2014 through 2019; $N=1,472$).¹⁰ In the late subperiod (R1b) the coefficient is -1.06 in compact districts and -1.43 in low-EC counties, both significant at the 5 percent level or better and well above the overall-sample magnitude. Of the two subperiods, the early one (R1a) is the row in which the overall-sample coefficient slips to insignificance, but the low-EC coefficient of -1.28 remains significant at the 10 percent level. Importantly, the early and late coefficients are of similar magnitudes and not statistically different from each other within any of the three samples, so the overall pattern is consistent with no structural change in the housing-inequality effect over the 11-year sample window rather than with attenuation that loads disproportionately on one subperiod.

Distributional decomposition. Row R2 of Table 6 decomposes the home-value 90/10 ratio into its upper half (the 90/50 ratio) and its lower half (the 50/10 ratio) and enters both variables into the regression simultaneously. The 90/10 ratio is mechanically the product of these two halves, so any decomposition of this kind tells us where in the housing distribution the inequality effect lives. The 50/10 coefficient is -0.82 in the compact-district only subsample and -1.16 in low-EC counties,

¹⁰The choice of break point reflects the unequal distribution of referenda across our sample window: referenda were more frequent in the years immediately following the Great Recession, and splitting at 2013 produces a more balanced division than splitting at the calendar midpoint.

both significant at the 5 percent level or better, while the 90/50 coefficient remains statistically indistinguishable from zero in every column. The negative effects of housing inequality therefore appear to be primarily driven by the disparity between the middle and the bottom of the district's housing stock rather than by the disparity between the middle and the top.¹¹ However, the adjusted R^2 is uniformly higher for R0 than R2, confirming that the parsimonious 90/10 specification carries the full explanatory content.

Alternative ACS merges. The main specification matches each referendum to a non-overlapping five-year ACS vintage that closes before the election year. Rows R3a and R3b of Table 6 report two alternative approaches. Row R3a applies the 2005–2009 ACS to every referendum in the sample, which fixes the demographic and housing controls at their pre-sample level and rules out any possible subsequent changes in those measures. Row R3b replaces the three-vintage merge with an annual lagged merge in which each referendum-year t uses the ACS five-year vintage closing at year $(t - 1)$ (with 2009 referenda matched contemporaneously to the 2005–2009 ACS, as that is the earliest 5-year ACS). Both alternative merges deliver materially larger coefficients in the compact and low-EC subsamples (R3a: -1.08 and -1.30 ; R3b: -1.03 and -1.26 , all significant at the 5 percent level or better). These estimates suggest our findings are robust to alternative assumptions about the appropriate temporal merge of demographic characteristics.

Spending purpose. Rows R4a and R4b of Table 6 split the sample by the purpose of the spending the referendum authorizes, estimating separately on capital referenda (R4a; $N=905$) and operating-expense referenda (R4b; $N=2,213$).¹² Capital referenda finance school construction, renovation, and major facility upgrades whose benefits are location-specific within a district; operating referenda fund teacher salaries, instructional materials, and transportation, whose benefits flow more immediately and more uniformly across the district. The visibility-of-housing-inequality mechanism makes a directional prediction here: the negative effect should be sharper for capital referenda than for operating-expense referenda, since a voter in a visibly unequal district has reason to ask whether a new facility will serve their own neighborhood or a wealthier part of the district. In the overall sample, the data line up with that prediction. Under the Full specification, the capital

¹¹Brunner and Ross (2010) provide complementary evidence on the location of the pivotal voter in school referenda. Using a pair of California ballot measures on the vote threshold for school bond passage, they find that the pivotal voter sits at approximately the 40th percentile of the income distribution rather than at the median. They interpret this as evidence of an ‘ends-against-the-middle’ coalition between low-income voters and relatively high-income voters with weak preferences for school spending. A pivotal voter located in the lower-middle of the income distribution would plausibly be more sensitive to housing-value dispersion between the bottom and middle of the local distribution than to dispersion at the top, consistent with the lower-half concentration of the effect we document here.

¹²Mixed-purpose referenda, which bundle capital and operating components within a single ballot question and do not admit a clean cross-type prediction, are dropped from this split.

coefficient is -0.72 and significant at the 5 percent level, while the operating-expense coefficient is -0.28 and statistically indistinguishable from zero. In the compact subsample, these patterns are mirrored with the coefficient for capital referenda (-1.33 ; p -value = 0.025) being nearly twice as large as the coefficient for operating-expense referenda (-0.69 ; p -value = 0.224). However, the ordering of magnitudes is opposite for the low-EC sample, though both referenda types have large, significant effects.

Election type. Rows R5a and R5b of Table 6 split the sample by election type. Within general elections (R5a; $N=858$), the home-value 90/10 coefficient is -0.64 in the overall sample, -0.81 in compact districts, and -0.14 in low-EC counties, all of which are statistically insignificant. Within non-general elections (R5b; $N=2,415$), the coefficients are -0.40 , -1.10 , and -1.69 across the same three samples, with the last two significant at the 5 and 1 percent level, respectively. While the corresponding coefficients between R5a and R5b are not statistically different, the magnitudes are substantially larger for non-general elections in the compact and low-EC subsamples, particularly low-EC. As stated above, non-general elections have lower turnout, and the voters who do turn out tend to be more engaged and have more instrumental value in the outcome (Kogan et al., 2018). These results are suggestive that, particularly in low-EC districts, the engaged voters showing up in non-general elections are also more aware of and willing to vote based on housing inequality.

Multiple referenda on the same ballot or in the same year. The headline specification treats each referendum as a separate observation, even when a district holds more than one referendum on the same election date or across multiple ballot dates within a single calendar year. This practice over-weights districts that go to the ballot frequently. Rows R6a and R6b of Table 6 address this over-weighting by restricting the sample to a single referendum per district per time window. Row R6a keeps only the referendum with the highest total turnout within each district-ballot-date combination (dropping 186 observations),¹³ and row R6b keeps only the referendum with the highest total turnout within each district-calendar-year combination (dropping 729 observations).¹⁴ Under both restrictions, the home-value 90/10 coefficient is negative and significant across all three samples (R6a: -0.47 , -1.06 , -1.38 ; R6b: -0.54 , -1.01 , -1.29). The point estimates are somewhat larger in magnitude than the headline R0 result, suggesting that any bias in the main specification from counting multiple referenda in the same district is, if anything, attenuating the effect.

¹³This choice allows a district to show up twice if it has a referendum on the ballot in the May primary elections and in the November elections.

¹⁴The motivation for selecting on highest turnout is that it tracks the question that drew the most engagement and therefore best reflects the median voter's preferences within that time window.

5 Conclusion

Voters respond to the inequality they can directly observe, not to the statistical income distribution that constitutes the standard object of study. We document this in a setting where voter choices have real fiscal stakes: school district referenda in Ohio, where local property tax authority cannot grow with inflation unless voters return to the ballot and approve it. Across 3,273 referenda between 2009 and 2019, the within-district 90/10 ratio of home values, a quantity that voters encounter in their physical environment and through online platforms such as Zillow and Realtor.com, predicts the yes-vote share. The Gini coefficient of household income, which voters cannot directly observe and which most cannot accurately characterize, does not.

The strength of the perceptions-based reading rests heavily on two heterogeneity results that go beyond establishing the average effect. First, the negative effect of visible housing inequality is concentrated in geographically compact districts and is essentially absent in geographically large ones. In districts below the sample median land area, the home-value 90/10 coefficient is approximately twice the pooled estimate and statistically significant at the 5 percent level; in larger districts, the coefficient is statistically indistinguishable from zero. Second, the effect is concentrated in counties with weak cross-income social ties as measured by the Chetty et al. (2022) economic connectedness index, and is substantially attenuated in counties where such ties are strong. The cross-EC slope difference is significant at the 1 percent level, with the binary-indicator interaction (+1.66) larger in absolute value than the home-value 90/10 main effect in low-EC counties (−1.34), implying that the negative effect of inequality is completely offset in counties where cross-income social ties are stronger. These two moderations matter not only because they sharpen the perceptions interpretation, but because they constitute a joint test against omitted variable bias explanations. A confounder correlated with the home-value 90/10 ratio and the yes-vote share, but operating through a channel other than visibility and social ties, would have to be systematically correlated with both district land area and county-level economic connectedness, and in exactly the right direction in each case. The visibility-and-social-cues mechanism delivers both patterns through a single channel: housing inequality is salient where the housing stock is observable to voters, and the political response to that salience is dampened where cross-income social ties knit residents of different income levels together. We are not aware of a standard omitted-variable story that would predict the joint pattern as readily.

Our findings extend the behavioral and political-economy literature on perceived inequality. The lab and survey-based work that has dominated this literature, including Choi (2019),

Domènech-Arú (2025), Gimpelson and Treisman (2018), and Norton and Ariely (2011), establishes that perceived inequality diverges from measured inequality and predicts stated policy preferences. Our results show that this gap survives the transition from stated preferences to revealed political behavior in a setting with concrete fiscal stakes, and that the gap operates through channels of physical exposure and social structure. The moderation by economic connectedness in particular brings the literature on bridging social capital into direct contact with the literature on inequality perceptions: visible inequality becomes politically consequential in social environments where cross-income contact is rare.

While this study establishes that visible housing inequality reduces referendum support, it cannot identify the precise psychological or social pathway. Several mechanisms are consistent with the evidence. Visible housing inequality may signal that the district is divided into socioeconomic groups with conflicting interests, weakening the shared sense of community that supports collective investment. Alternatively, voters in unequal districts may be skeptical that funding increases will be distributed equitably across neighborhoods. A third possibility, consistent with Slemrod (2006), is that visible inequality shapes perceptions of who bears the tax burden, leading voters to misidentify whether a referendum is progressive or regressive in practice. Distinguishing among these channels would require survey or experimental data that lie beyond the scope of this study.

Several avenues for future research follow. Survey and experimental work would help identify whether social fragmentation or benefit distrust is the dominant channel through which visible inequality depresses collective fiscal support. Comparative research across states with different school finance systems would clarify whether the inequality effect generalizes beyond Ohio. Extending the analysis to other local public goods such as parks, libraries, and public safety would test whether visible inequality shapes support for a broader range of collective investments. Should those extensions confirm the pattern documented here, the broader implication is that the inequality voters perceive in their environment, rather than the inequality social scientists measure, is what shapes the political feasibility of collective investment.

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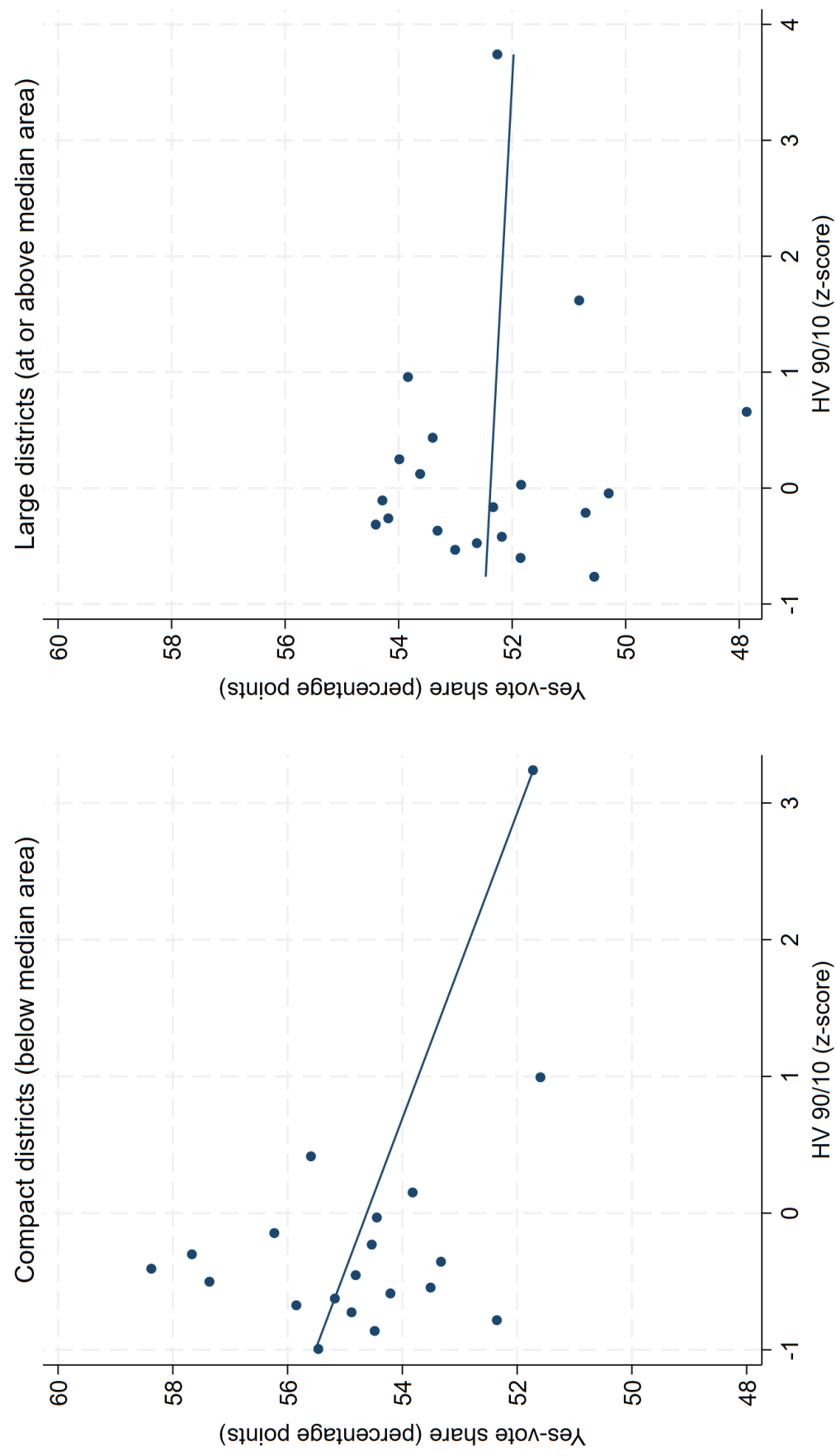
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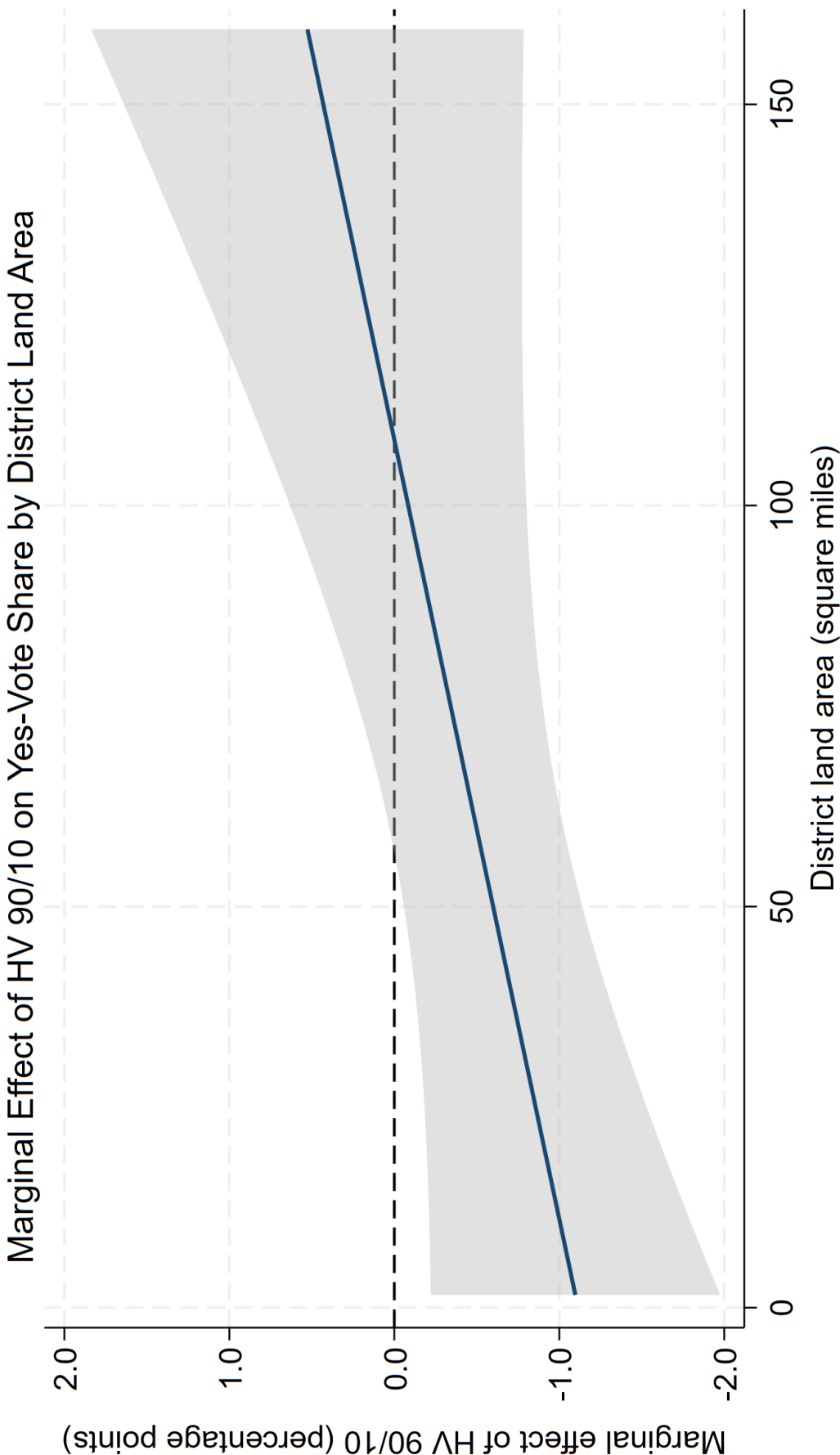
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Figure 1: Yes-vote share against HV 90/10, by district size



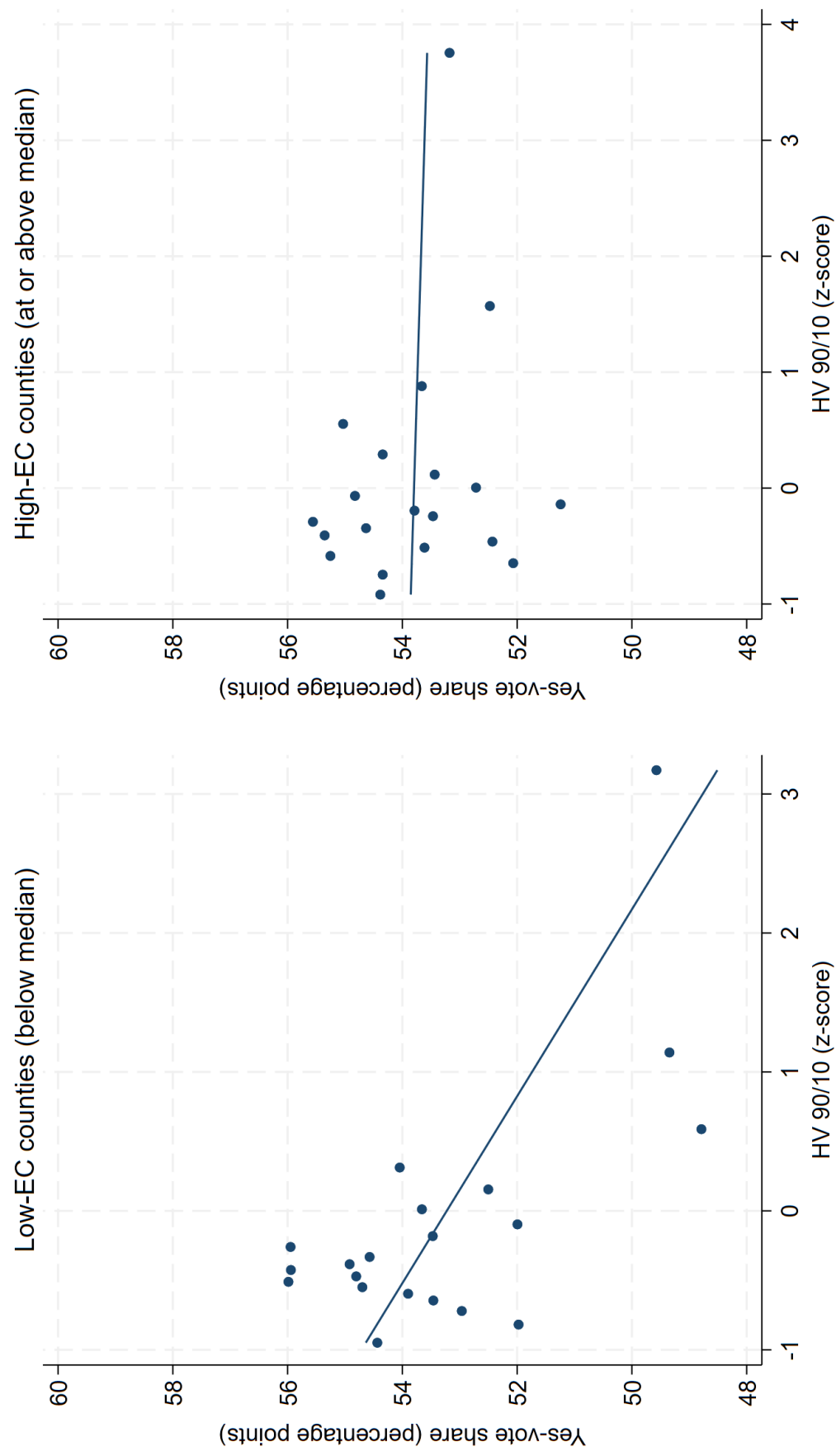
Notes: Figure presents binscatters of the residualized yes-vote share against the z-scored home-value 90/10 ratio, separately for compact districts (those with land area below the sample median of approximately 48 square miles) and large districts (at or above the median). The figure is constructed using the binscatter package (Stepner, 2013) with 20 equal-sized bins per panel, residualizing on election-year fixed effects. The negative slope visible in the left panel is the visual analog of the split-sample regression results in Columns 1 and 2 of Table 3.

Figure 2: Marginal effect of HV 90/10 on the yes-vote share, by district land area



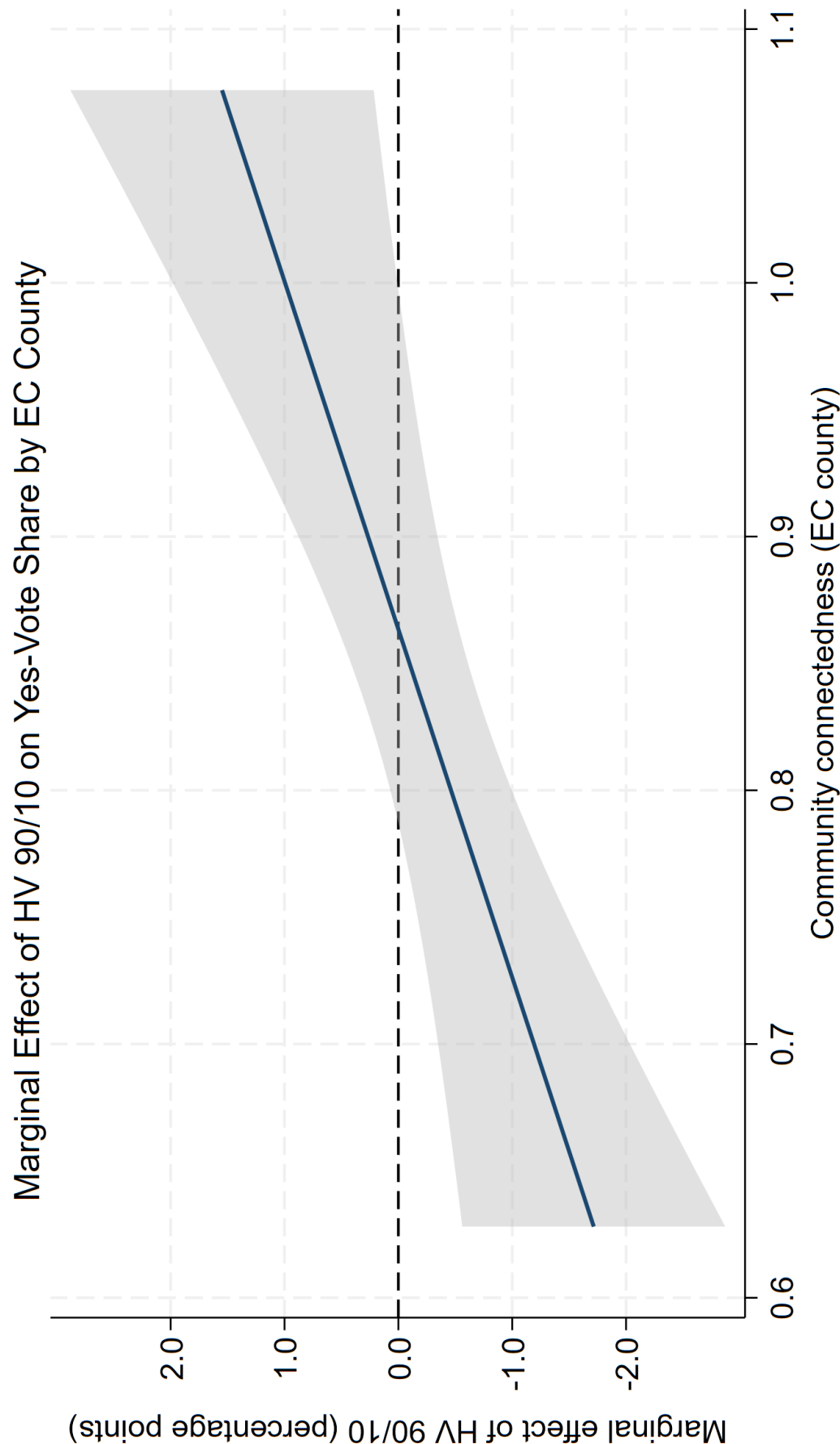
Notes: Figure plots the marginal effect of the home-value 90/10 ratio (z-scored) on the yes-vote share as a function of district land area, estimated from a fully interacted continuous-area specification (the binary-area analog of this exercise appears in Column 3 of Table 3). The shaded region is the 95% confidence interval. Standard errors are clustered at the school district level. The x-axis is capped at the 95th percentile of the area distribution so the plot is not dominated by a small number of very large districts.

Figure 3: Yes-vote share against HV 90/10, by economic connectedness



Notes: Figure presents bincatters of the residualized yes-vote share against the z-scored home-value 90/10 ratio, separately for low-EC counties (those with county-level economic connectedness below the sample median) and high-EC counties (at or above the median). The figure is constructed using the bincscatter package (Stepner, 2013) with 20 equal-sized bins per panel, residualizing on election-year fixed effects. The pronounced negative slope in the left panel is the visual analog of the split-sample regression results in Columns 1 and 2 of Table 4 Panel A.

Figure 4: Marginal effect of HV 90/10 on the yes-vote share, by economic connectedness



Notes: Figure plots the marginal effect of the home-value 90/10 ratio (z-scored) on the yes-vote share as a function of county-level economic connectedness, estimated from a fully interacted continuous-EC specification (the binary-EC analog of this exercise appears in Column 3 of Table 4 Panel A). The shaded region is the 95% confidence interval. Standard errors are clustered at the school district level. Higher values of economic connectedness indicate greater cross-income social integration in the county.

Table 1: Summary statistics

	Mean	SD	Median	Min	Max
Yes-vote share (%)	53.6	11.0	53.9	14.0	87.3
<i>Housing inequality</i>					
Home-value 90/10 ratio	5.692	3.224	4.809	2.435	35.855
Home-value 90/10 (winsorized 1%/99%)	5.652	2.965	4.809	2.706	20.540
Home-value 90/50 ratio	2.129	0.309	2.089	1.377	3.460
Home-value 50/10 ratio	2.621	1.230	2.255	1.365	12.263
<i>Other inequality, community structure</i>					
Gini coefficient (household income)	0.399	0.040	0.395	0.292	0.557
Ethnic fractionalization (%)	14.5	12.5	10.2	0.0	66.4
Economic connectedness (county)	0.799	0.105	0.793	0.628	1.076
Residential integration	0.893	0.065	0.906	0.564	1.000
<i>Demographic and economic controls</i>					
Share aged under 18 (%)	23.4	3.6	23.3	10.2	41.4
Share aged 65 and older (%)	15.3	3.3	15.3	4.6	33.6
Homeownership rate (%)	76.5	9.8	78.3	36.4	96.3
Share with bachelor's degree (%)	21.0	12.1	17.1	3.9	75.5
Democratic share, 2012 (%)	46.8	11.3	45.8	21.9	69.4
Mean home value (USD)	170,932	50,826	163,018	63,246	482,162
Median household income (USD)	53,775	13,248	52,026	24,825	120,931
Population density (per sq. mi.)	704	1,036	254	25	9,420
District enrollment (students)	2,702	3,372	1,812	94	50,488
District land area (sq. mi.)	62	58	48	2	543
<i>Referendum purpose</i>					
Capital	0.277	0.447	0	0	1
Operations	0.676	0.468	1	0	1
Mixed-purpose	0.047	0.212	0	0	1
<i>Election timing</i>					
General (November of even year)	0.262	0.440	0	0	1
Non-general (primaries, specials, odd-year November)	0.738	0.440	1	0	1
Observations: 3,273					

Rates and shares are reported in percentage points. Mean home value, median household income, population density, district enrollment, and district land area are reported in their native units. Housing-inequality ratios, Gini, economic connectedness, and residential integration are reported on their native scales (0–1 for the Gini, residential integration, and economic connectedness; ratio for the housing-value measures).

Table 2: Main results: visible housing inequality and the yes-vote share

	(1)	(2)	(3)	(4)	(5)
HV 90/10 (z-score)	-0.682** (0.289)	-0.746** (0.294)	-0.649** (0.302)	-0.461* (0.270)	-0.448* (0.264)
Gini coefficient		18.849*** (7.288)	17.257** (7.383)	-6.794 (8.943)	-6.316 (8.985)
Ethnic fractionalization		1.626 (2.676)	2.254 (2.690)	1.036 (3.445)	1.200 (3.470)
Log mean home value			1.164 (1.054)	-3.791* (2.145)	-4.111* (2.135)
Share aged under 18				18.037* (9.325)	19.418** (9.365)
Share aged 65 and older				7.903 (11.275)	7.891 (11.192)
Homeownership rate				-6.651 (5.425)	-7.208 (5.492)
Log median household income				-3.331 (3.303)	-3.098 (3.289)
Log population density				0.754* (0.399)	0.704* (0.400)
Share with bachelor's degree				29.758*** (4.915)	31.321*** (4.892)
Democratic share 2012				-14.747*** (2.884)	-14.488*** (2.923)
Log district enrollment				-2.393*** (0.526)	-2.417*** (0.530)
Mixed-purpose referendum					-4.256*** (1.142)
Operations referendum					0.725 (0.496)
Non-general election					0.438 (0.506)
N	3273	3273	3273	3273	3273
R-squared	0.056	0.061	0.062	0.115	0.124

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Dependent variable is the yes-vote share expressed in percentage points. The home-value 90/10 ratio is winsorized at the 1st and 99th percentiles and standardized to a z-score so the coefficient reads as the effect of a one-standard-deviation increase in visible housing inequality. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. Demographic controls in Columns (4) and (5) are the share of the district population under 18, the share aged 65 and older, the homeownership rate, log median household income, log population density, the share with a bachelor's degree, the Democratic share of the 2012 presidential vote at the county level, and log district enrollment. Election-specific controls in Column (5) are indicators for whether the referendum funds capital, operations, or a mix of the two, and an indicator for whether the election is a non-general election (i.e., one not held in November of an even-numbered year).

Table 3: Heterogeneity by district size

	(1)	(2)	(3)
HV 90/10 (z-score)	-0.992** (0.447)	0.052 (0.319)	0.058 (0.320)
Gini coefficient	2.032 (12.439)	-20.524 (13.020)	-20.172 (13.112)
HV 90/10 $\times$ Small district			-1.082** (0.546)
Small district (below median area)			-73.484 (55.266)
Gini $\times$ Small district			21.894 (17.811)
Gini and ethnic fractionalization	Yes	Yes	Yes
Log mean home value	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Election-specific controls	Yes	Yes	Yes
N	1639	1634	3273
R-squared	0.114	0.156	0.144

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Dependent variable is the yes-vote share expressed in percentage points. Compact districts are those with land area below the sample median (approximately 48 square miles); large districts are at or above the median. Column (1) reports the regression estimated on the compact-district subsample; column (2) reports the parallel estimate on the large-district subsample; column (3) reports a pooled fully interacted specification in which every control is allowed to vary with the compact-district indicator, so the coefficient on the interaction of the home-value 90/10 ratio with the compact-district indicator is the cross-subsample slope difference without imposing any common-control restriction. All three columns include the full set of demographic, economic, and political controls along with the spending-category and election-type indicators. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. Demographic controls and election-specific controls are as defined in the notes to Table 2.

Table 4: Heterogeneity by community structure

Panel A: Economic connectedness			
	(1)	(2)	(3)
HV 90/10 (z-score)	-1.370*** (0.495)	0.404 (0.298)	-1.341*** (0.476)
HV 90/10 $\times$ High-EC county			1.657*** (0.558)
High-EC county (above median)			-56.455 (57.977)
Gini and ethnic fractionalization	Yes	Yes	Yes
Log mean home value	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Election-specific controls	Yes	Yes	Yes
N	1621	1652	3273
R-squared	0.115	0.182	0.146
Standard errors in parentheses			
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$			
Panel B: Residential integration			
	(1)	(2)	(3)
HV 90/10 (z-score)	-0.598 (0.457)	-0.403 (0.286)	3.064 (3.669)
HV 90/10 $\times$ Residential integration			-3.827 (3.983)
Residential integration			343.964 (399.140)
Gini and ethnic fractionalization	Yes	Yes	Yes
Log mean home value	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Election-specific controls	Yes	Yes	Yes
N	1632	1641	3273
R-squared	0.118	0.159	0.138
Standard errors in parentheses			
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$			

Notes: Dependent variable is the yes-vote share expressed in percentage points. Panel A splits the sample at the median of the county-level Chetty et al. (2022) economic-connectedness index; Panel B splits the sample at the median of the residential integration measure (one minus Theil's H across census tracts within each district). Within each panel, column (1) reports the regression estimated on the low-moderator subsample (below median), column (2) reports the parallel estimate on the high-moderator subsample (at or above median), and column (3) reports a pooled fully interacted specification: in Panel A, every control is allowed to vary with a binary indicator for above-median economic connectedness; in Panel B, every control is allowed to vary continuously with the residential-integration measure. The coefficient on the interaction of the home-value 90/10 ratio with the moderator is the cross-subsample slope difference without imposing any common-control restriction. All three columns include the full set of demographic, economic, and political controls along with the spending-category and election-type indicators. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. Demographic controls and election-specific controls are as defined in the notes to Table 2.

Table 5: Joint moderation: district size and economic connectedness

	(1)	(2)	(3)	(4)	(5)
HV 90/10 (z-score)	-0.769*	-0.653	-0.758	-0.927*	-0.930*
	(0.436)	(0.457)	(0.484)	(0.486)	(0.502)
HV 90/10 $\times$ Small district	-0.792	-1.150**	-0.874	-0.848	-0.742
	(0.541)	(0.561)	(0.567)	(0.525)	(0.522)
HV 90/10 $\times$ High-EC county	1.111**	1.190**	1.090*	1.549***	1.481***
	(0.535)	(0.554)	(0.567)	(0.538)	(0.541)
Gini and ethnic fractionalization	No	Yes	Yes	Yes	Yes
Log mean home value	No	No	Yes	Yes	Yes
Demographic controls	No	No	No	Yes	Yes
Election-specific controls	No	No	No	No	Yes
N	3273	3273	3273	3273	3273
R-squared	0.073	0.085	0.088	0.148	0.159

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

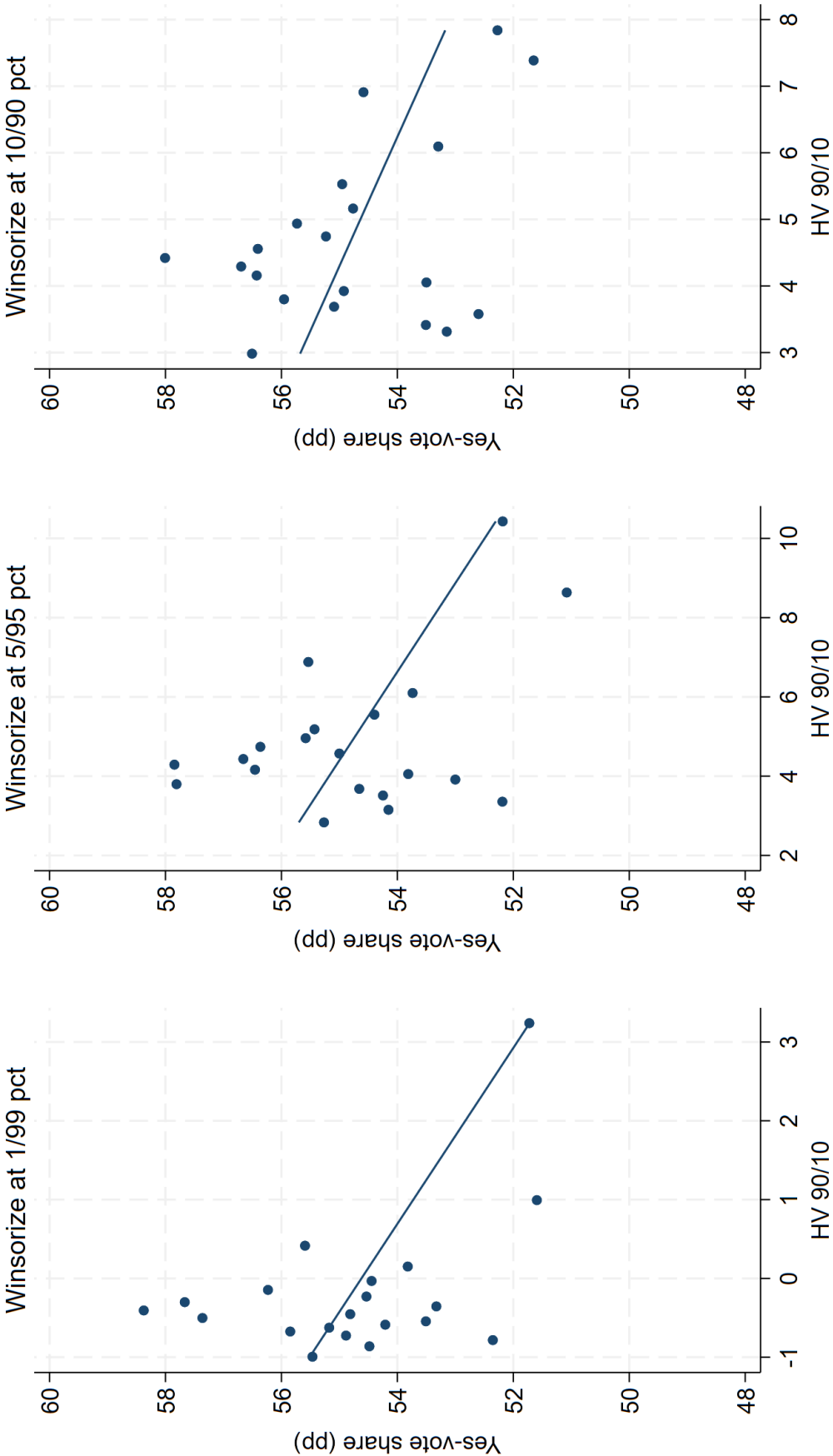
Notes: Dependent variable is the yes-vote share expressed in percentage points. The table repeats the column progression of Table 2 with both interactions — the home-value 90/10 ratio crossed with the compact-district indicator and with a binary indicator for above-median (high) economic connectedness — added to every specification. Each control present in a given column is fully interacted with both moderators. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. Demographic controls and election-specific controls are as defined in the notes to Table 2.

Table 6: Robustness checks across samples: HV 90/10 coefficient under the Full specification

	(1) Overall	(2) Compact	(3) Low-EC
R0: Baseline specification (HV 90/10)	-0.448* (0.264)	-0.992** (0.447)	-1.370*** (0.495)
R1a: Early subperiod (2009–2013)	-0.326 (0.405)	-0.834 (0.632)	-1.280* (0.754)
R1b: Late subperiod (2014–2019)	-0.511* (0.297)	-1.058** (0.502)	-1.432*** (0.455)
R2: Distributional decomposition (HV 50/10)	-0.406 (0.250)	-0.818** (0.398)	-1.156*** (0.425)
Distributional decomposition (HV 90/50)	0.103 (0.373)	-0.208 (0.530)	-0.345 (0.539)
R3a: Single 2005–2009 ACS vintage	-0.592** (0.270)	-1.082** (0.438)	-1.295*** (0.478)
R3b: Annual lagged ACS merge	-0.436 (0.276)	-1.030** (0.453)	-1.256*** (0.493)
R4a: Capital referenda only	-0.719** (0.343)	-1.328** (0.588)	-1.270** (0.559)
R4b: Operating-expense referenda only	-0.283 (0.364)	-0.691 (0.567)	-1.617** (0.652)
R5a: General elections only	-0.644 (0.397)	-0.814 (0.554)	-0.142 (0.574)
R5b: Non-general elections only	-0.399 (0.307)	-1.104** (0.553)	-1.692*** (0.600)
R6a: Highest-turnout referendum per ballot date	-0.469* (0.274)	-1.058** (0.430)	-1.380** (0.539)
R6b: Highest-turnout referendum per calendar year	-0.536* (0.279)	-1.007** (0.455)	-1.288** (0.552)

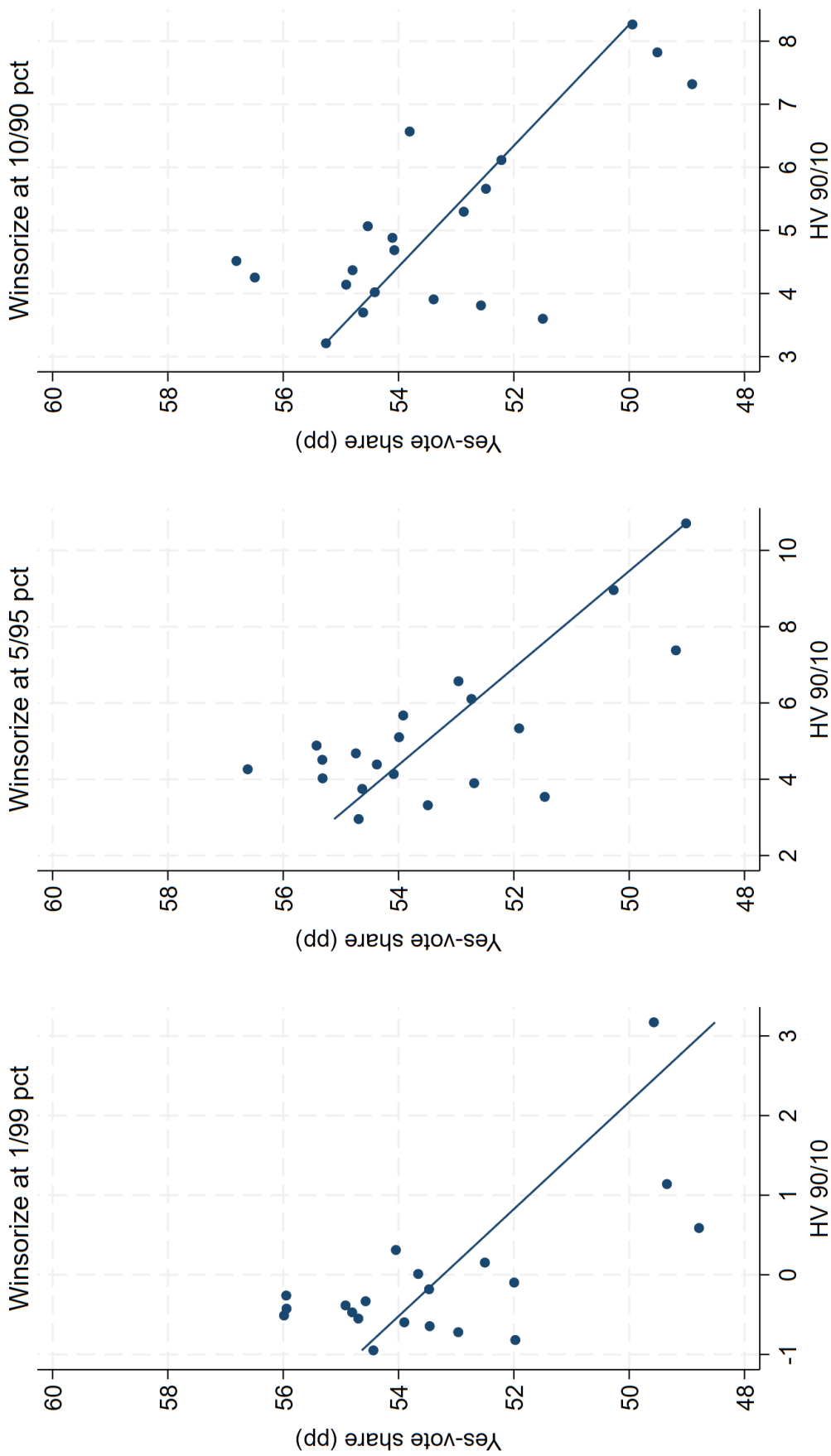
Notes: Dependent variable is the yes-vote share expressed in percentage points. Each cell reports the home-value 90/10 coefficient (and, for row R2, the coefficients on the home-value 50/10 and 90/50 ratios) from the Full specification (column 5 of Table 2) estimated separately on three samples: (1) the overall sample, (2) compact districts only (below median land area), and (3) low-EC counties only (below median county-level economic connectedness). Each row corresponds to a robustness specification described in Section 4.5; the full column progression for each sample appears in Appendix Tables A.2, A.3, and A.4. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses.

Figure A.1: Compact-district binscatter under alternative winsorization



Notes: Figure presents binscatter plots of the residualized yes-vote share against the home-value 90/10 ratio for compact districts (those with land area below the sample median) under three alternative winsorization choices: 1st/99th percentiles (the main-text specification), 5th/95th, and 10th/90th. The figure is constructed using the binscatter package (Stepner, 2013) with 20 equal-sized bins. Standard errors are clustered at the school district level.

Figure A.2: Low-EC bincscatter under alternative winsorization



Notes: Figure presents bincscatter plots of the residualized yes-vote share against the home-value 90/10 ratio for districts in low-EC counties (those with county-level economic connectedness below the sample median) under three alternative winsorization choices: 1st/99th percentiles (the main-text specification), 5th/95th, and 10th/90th. The figure is constructed using the bincscatter package (Stepner, 2013) with 20 equal-sized bins. Standard errors are clustered at the school district level.

Table A.1: Pairwise correlations: inequality measures and moderators

	HV 90/10	Gini	Eth. frac.	Mean HV	Small district	EC (county)	Res. integ.
Home-value 90/10 ratio	1.000						
Gini coefficient	0.140***	1.000					
Ethnic fractionalization	-0.134***	0.328***	1.000				
Mean home value	-0.212***	0.152***	-0.083***	1.000			
Small district (below median area)	-0.166***	0.190***	0.294***	0.101***	1.000		
Economic connectedness (county)	0.009	-0.179***	-0.238***	0.206***	-0.259***	1.000	
Residential integration	0.039**	-0.127***	-0.180***	0.113***	-0.063***	0.239***	1.000
N	3273						

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: Robustness checks: overall sample

	(1)	(2)	(3)	(4)	(5)
R0: Baseline specification (HV 90/10)	-0.682** (0.289)	-0.746** (0.294)	-0.649** (0.302)	-0.461* (0.270)	-0.448* (0.264)
R1a: Early subperiod (2009–2013)	-0.696* (0.376)	-0.786** (0.380)	-0.538 (0.388)	-0.335 (0.422)	-0.326 (0.405)
R1b: Late subperiod (2014–2019)	-0.670* (0.344)	-0.711** (0.352)	-0.726** (0.366)	-0.547* (0.312)	-0.511* (0.297)
R2: Distributional decomposition (HV 50/10)	-0.653** (0.296)	-0.652** (0.294)	-0.615** (0.294)	-0.406 (0.255)	-0.406 (0.250)
Distributional decomposition (HV 90/50)	-0.001 (0.328)	-0.123 (0.332)	0.071 (0.354)	0.075 (0.371)	0.103 (0.373)
R3a: Single 2005–2009 ACS vintage	-0.711** (0.281)	-0.828*** (0.285)	-0.764** (0.304)	-0.604** (0.278)	-0.592** (0.270)
R3b: Annual lagged ACS merge	-0.696** (0.293)	-0.764** (0.301)	-0.657** (0.312)	-0.461 (0.282)	-0.436 (0.276)
R4a: Capital referenda only	-1.123*** (0.367)	-1.186*** (0.371)	-1.066*** (0.381)	-0.719** (0.345)	-0.719** (0.343)
R4b: Operating-expense referenda only	-0.411 (0.368)	-0.473 (0.376)	-0.414 (0.389)	-0.261 (0.365)	-0.283 (0.364)
R5a: General elections only	-0.878** (0.346)	-0.965*** (0.361)	-0.916** (0.382)	-0.680* (0.395)	-0.644 (0.397)
R5b: Non-general elections only	-0.623* (0.345)	-0.689* (0.355)	-0.573 (0.365)	-0.400 (0.314)	-0.399 (0.307)
R6a: Highest-turnout referendum per ballot date	-0.733** (0.285)	-0.810*** (0.289)	-0.693** (0.298)	-0.487* (0.277)	-0.469* (0.274)
R6b: Highest-turnout referendum per calendar year	-0.863*** (0.291)	-0.883*** (0.298)	-0.773** (0.308)	-0.539* (0.291)	-0.536* (0.279)
Gini and ethnic fractionalization	No	Yes	Yes	Yes	Yes
Log mean home value	No	No	Yes	Yes	Yes
Demographic controls	No	No	No	Yes	Yes
Election-specific controls	No	No	No	No	Yes

Notes: Dependent variable is the yes-vote share expressed in percentage points. Each row reports the home-value 90/10 coefficient (and, for row R2, the coefficients on the home-value 50/10 and 90/50 ratios) from a separate regression that mirrors the column progression of Table 2. Row R2 enters the home-value 50/10 ratio and the home-value 90/50 ratio jointly within the same regression. R6a restricts the sample to the referendum with the highest total turnout within each district-ballot-date combination; R6b restricts to the referendum with the highest total turnout within each district-calendar-year combination. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. The Yes/No rows at the bottom indicate which control groups are included in each column; control groups are defined as in the notes to Table 2.

Table A.3: Robustness checks: compact-district subsample

	(1)	(2)	(3)	(4)	(5)
R0: Baseline specification (HV 90/10)	-0.896** (0.412)	-1.126*** (0.422)	-0.945** (0.414)	-1.047** (0.458)	-0.992** (0.447)
R1a: Early subperiod (2009–2013)	-1.118** (0.560)	-1.375** (0.544)	-1.048* (0.558)	-0.833 (0.658)	-0.834 (0.632)
R1b: Late subperiod (2014–2019)	-0.683 (0.456)	-0.887* (0.473)	-0.837* (0.473)	-1.127** (0.530)	-1.058** (0.502)
R2: Distributional decomposition (HV 50/10)	-0.885** (0.369)	-0.929** (0.374)	-0.853** (0.375)	-0.865** (0.407)	-0.818** (0.398)
Distributional decomposition (HV 90/50)	0.049 (0.505)	-0.307 (0.512)	-0.060 (0.509)	-0.233 (0.528)	-0.208 (0.530)
R3a: Single 2005–2009 ACS vintage	-0.904** (0.377)	-1.166*** (0.372)	-1.017*** (0.377)	-1.129** (0.452)	-1.082** (0.438)
R3b: Annual lagged ACS merge	-0.925** (0.408)	-1.154*** (0.432)	-0.948** (0.435)	-1.083** (0.460)	-1.030** (0.453)
R4a: Capital referenda only	-1.344** (0.590)	-1.675*** (0.615)	-1.483** (0.625)	-1.325** (0.588)	-1.328** (0.588)
R4b: Operating-expense referenda only	-0.580 (0.477)	-0.793* (0.477)	-0.666 (0.463)	-0.697 (0.567)	-0.691 (0.567)
R5a: General elections only	-0.934* (0.484)	-1.082** (0.504)	-0.907* (0.515)	-0.848 (0.550)	-0.814 (0.554)
R5b: Non-general elections only	-0.901* (0.485)	-1.177** (0.488)	-0.991** (0.483)	-1.155** (0.562)	-1.104** (0.553)
R6a: Highest-turnout referendum per ballot date	-0.928** (0.400)	-1.180*** (0.406)	-1.003** (0.405)	-1.113** (0.438)	-1.058** (0.430)
R6b: Highest-turnout referendum per calendar year	-0.948** (0.416)	-1.139*** (0.431)	-0.978** (0.438)	-1.138** (0.474)	-1.007** (0.455)
Gini and ethnic fractionalization	No	Yes	Yes	Yes	Yes
Log mean home value	No	No	Yes	Yes	Yes
Demographic controls	No	No	No	Yes	Yes
Election-specific controls	No	No	No	No	Yes

Notes: Dependent variable is the yes-vote share expressed in percentage points. The sample is restricted to compact districts (below median land area). Each row reports the home-value 90/10 coefficient (and, for row R2, the coefficients on the home-value 50/10 and 90/50 ratios) from a separate regression that mirrors the column progression of Table 2. Row R2 enters the home-value 50/10 ratio and the home-value 90/50 ratio jointly within the same regression. R6a restricts the sample to the referendum with the highest total turnout within each district-ballot-date combination; R6b restricts to the referendum with the highest total turnout within each district-calendar-year combination. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. The Yes/No rows at the bottom indicate which control groups are included in each column; control groups are defined as in the notes to Table 2.

Table A.4: Robustness checks: low-EC subsample

	(1)	(2)	(3)	(4)	(5)
R0: Baseline specification (HV 90/10)	-1.487*** (0.439)	-1.644*** (0.450)	-1.491*** (0.458)	-1.419*** (0.494)	-1.370*** (0.495)
R1a: Early subperiod (2009–2013)	-1.441** (0.577)	-1.637*** (0.577)	-1.342** (0.596)	-1.273* (0.767)	-1.280* (0.754)
R1b: Late subperiod (2014–2019)	-1.536*** (0.481)	-1.631*** (0.482)	-1.617*** (0.499)	-1.589*** (0.483)	-1.432*** (0.455)
R2: Distributional decomposition (HV 50/10)	-1.442*** (0.403)	-1.431*** (0.406)	-1.363*** (0.409)	-1.205*** (0.421)	-1.156*** (0.425)
Distributional decomposition (HV 90/50)	-0.116 (0.470)	-0.439 (0.455)	-0.228 (0.484)	-0.343 (0.535)	-0.345 (0.539)
R3a: Single 2005–2009 ACS vintage	-1.450*** (0.388)	-1.572*** (0.401)	-1.410*** (0.418)	-1.340*** (0.476)	-1.295*** (0.478)
R3b: Annual lagged ACS merge	-1.368*** (0.431)	-1.562*** (0.460)	-1.410*** (0.480)	-1.317*** (0.491)	-1.256** (0.493)
R4a: Capital referenda only	-1.665*** (0.557)	-1.918*** (0.551)	-1.698*** (0.575)	-1.288** (0.569)	-1.270** (0.559)
R4b: Operating-expense referenda only	-1.394** (0.547)	-1.589*** (0.559)	-1.519*** (0.579)	-1.611** (0.651)	-1.617** (0.652)
R5a: General elections only	-0.662 (0.475)	-0.743 (0.488)	-0.541 (0.511)	-0.173 (0.584)	-0.142 (0.574)
R5b: Non-general elections only	-1.754*** (0.500)	-1.953*** (0.517)	-1.818*** (0.524)	-1.768*** (0.587)	-1.692*** (0.600)
R6a: Highest-turnout referendum per ballot date	-1.483*** (0.453)	-1.655*** (0.461)	-1.488*** (0.473)	-1.421*** (0.534)	-1.380** (0.539)
R6b: Highest-turnout referendum per calendar year	-1.552*** (0.495)	-1.636*** (0.503)	-1.499*** (0.522)	-1.333** (0.567)	-1.288** (0.552)
Gini and ethnic fractionalization	No	Yes	Yes	Yes	Yes
Log mean home value	No	No	Yes	Yes	Yes
Demographic controls	No	No	No	Yes	Yes
Election-specific controls	No	No	No	No	Yes

Notes: Dependent variable is the yes-vote share expressed in percentage points. The sample is restricted to low-EC counties (below median county-level economic connectedness). Each row reports the home-value 90/10 coefficient (and, for row R2, the coefficients on the home-value 50/10 and 90/50 ratios) from a separate regression that mirrors the column progression of Table 2. Row R2 enters the home-value 50/10 ratio and the home-value 90/50 ratio jointly within the same regression. R6a restricts the sample to the referendum with the highest total turnout within each district-ballot-date combination; R6b restricts to the referendum with the highest total turnout within each district-calendar-year combination. All specifications include election-year fixed effects. Standard errors clustered at the school district level are reported in parentheses. The Yes/No rows at the bottom indicate which control groups are included in each column; control groups are defined as in the notes to Table 2.